Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — qualifier level (Listas 1–2)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — Price indices, new goods, welfare and growth accounting

(Kurlat chs. 1–2, 5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household with CES preferences (elasticity of substitution $\eta > 0$, $\eta \neq 1$) over two goods spends a share $\omega \in (0,1)$ of its budget on good 1 and $1-\omega$ on good 2 at base prices. The price of good 1 is multiplied by $R > 1$; the price of good 2 does not change. The log overstatement of the true change in the cost of living by the Laspeyres index L, $\ln(L/P^*)$ with P^* the exact index, is:

A. $\ln(\omega R + 1 - \omega) - \omega \ln R$

B. $\ln(\omega R + 1 - \omega) - \frac{1}{\eta-1}\ln(\omega R^{\eta-1} + 1 - \omega)$

C. $\ln(\omega R + 1 - \omega) - \frac{1}{1-\eta}\ln(\omega R^{1-\eta} + 1 - \omega)$

D. $\ln(\omega R + 1 - \omega) - \ln(\omega R^{\eta-1} + 1 - \omega)$

b) (3 points) *Welfare × growth.* Over T years Brazil's mean consumption grows at g log points a year (initial mean one), the variance of log consumption (normal) goes from v_0 to v_T, and the survival probability of Kurlat's single-year experiment (life expectancy over 100) rises from a_0 to $a_T > a_0$; leisure is constant (absorbed into $\bar{u}$). Utility is Kurlat–Jones–Klenow with $\sigma = 1$ and positive flow utility at both dates. If λ scales all date-0 consumption so that Rawls is indifferent between date-0 and date-T Brazil, the average annual growth of consumption-equivalent welfare, $(\ln \lambda)/T$, is:

A. $g + \frac{v_0 - v_T}{2T} + \frac{1}{T}\left(\frac{a_T}{a_0} - 1\right)\left(\bar{u} + gT - \frac{v_T}{2}\right)$

B. $g + \frac{v_T - v_0}{2T} + \frac{1}{T}\left(\frac{a_T}{a_0} - 1\right)\left(\bar{u} + gT + \frac{v_T}{2}\right)$

C. $g + \frac{v_0 - v_T}{2T} + \frac{1}{T}\ln\frac{a_T}{a_0}$

D. $g + \frac{v_T - v_0}{2T} + \frac{1}{T}\left(1 - \frac{a_0}{a_T}\right)\left(\bar{u} + \frac{v_0}{2}\right)$

c) (2 points, T or F) A Paasche price index can never exceed the corresponding Laspeyres index.

d) (2 points, T or F) *PPP × productivity growth.* In Brazil and in the US labor is the only input and moves freely between tradables and non-tradables (linear technologies); consumers buy a Cobb–Douglas basket with the same non-tradable share $\gamma \in (0,1)$; tradables obey the law of one price; the exchange rate is constant. If Brazil's tradable productivity grows faster than the US's while non-tradable productivity grows equally in both, Brazil's price level relative to the US's rises, at a rate that increases with γ and would be zero if all goods were tradable.

Question 2. (15 points) Discursive. *Measurement × growth accounting.* Consumers have CES preferences over the set Ω_t of varieties available at date t,

$$U = \Big(\sum_{i \in \Omega_t} (b_i x_i)^{\frac{\eta-1}{\eta}} \Big)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1,$$

where $b_i > 0$ is a quality weight. The statistical agency (IBGE) observes prices and expenditure shares and builds its deflators from varieties present in both periods. Growth rates are in log points per period.

a) (3 points) Find the minimum expenditure needed to obtain one unit of utility at prices p, and each variety's expenditure share.

Solution:

b) (3 points) Keep the set of varieties fixed between dates 0 and 1. Express the exact index P_1/P_0 in terms of observable price relatives $R_i = p_{i1}/p_{i0}$ and shares, rank it against the Laspeyres and Paasche indices, and approximate the Laspeyres bias to second order in the dispersion of $\ln R_i$.

Solution:

c) (3 points) Now some varieties are new at date 1 and none disappears. Find the exact index in terms of the index over continuing varieties and observable shares. What happens as $\eta \to \infty$ and as $\eta \to 1$?

Solution:

d) (3 points) The agency measures the continuing-variety consumption index exactly. New

varieties appear only among consumption goods, take a share $\nu \in (0,1)$ of date-1 consumption spending, and none disappears. Investment goods (nominal GDP share $s_I \in (0,1)$, $s_C = 1 - s_I$) have their own deflator bias $\beta_I \in [0, g_K + \delta)$ from unmeasured quality change (g_K: true capital growth; $\delta > 0$: depreciation); labor and the capital share $\alpha \in (0,1)$ are measured correctly. On a balanced growth path, find the consumption-deflator bias β_C, show that measured capital growth converges and at what speed, and find the distortion $\Delta \equiv g_A^m - g_A$ of the measured Solow residual; compare $\beta_I = 0$ with $\beta_I = \beta_C$.

Solution:

e) (3 points) *Parameter analysis.* How do β_C and Δ depend on η, ν, β_I, α and s_I? Prove and tabulate the signs. Find the β_I^* at which Δ changes sign, when it exists, and how it moves with the parameters. Give the limits of β_C as $\eta \to 1^+$, $\eta \to \infty$ and $\nu \to 0$, and draw β_C against η for three new-variety shares $\nu_1 < \nu_2 < \nu_3$.

Solution:

Block 2 — Growth, capital flows and national income

(Kurlat chs. 4–5; Romer chs. 1, 4 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Development accounting × capital flows.* Let $Y = AK^{\alpha}(hL)^{1-\alpha}$, $\alpha \in (0,1)$, with the same A in Brazil and the US. Brazil's output per worker is a fraction ϕ of the US's and its human capital per worker a fraction χ, with $0 < \phi < \chi < 1$. The ratio of marginal products of capital (Brazil/US) implied by the closed-economy model is:

A. $(1/\phi)^{(1-\alpha)/\alpha}$

B. $(\chi/\phi)^{1-\alpha}$

C. $(1/\phi)^{1-\alpha}$

D. $(\chi/\phi)^{(1-\alpha)/\alpha}$

b) (3 points) *Growth × welfare.* Output per effective worker is $f(k) = A[\alpha k^{\psi} + 1 - \alpha]^{1/\psi}$ with $A > 0$, $\alpha \in (0,1)$, $\psi < 1$, $\psi \neq 0$ (elasticity of substitution $\sigma = 1/(1-\psi)$), and $\dot{k} = sf(k) - Nk$ with $N \equiv n + g + \delta > 0$. When a Golden-Rule capital stock exists, the saving rate that attains it is:

A. $(\alpha A^{\psi})^{1-\psi} N^{\psi}$

B. $(\alpha A^{\psi})^{-1/(1-\psi)} N^{\psi/(1-\psi)}$

C. $(\alpha A^{\psi})^{1/(1-\psi)} N^{-\psi/(1-\psi)}$

D. $(\alpha A^{\psi})^{-(1-\psi)} N^{-\psi}$

c) (2 points, T or F) Let $Y = [\alpha(BK)^{\psi} + (1-\alpha)(AL)^{\psi}]^{1/\psi}$, $\psi < 1$, $\psi \neq 0$, where B, A, L grow at the constant rates $g_B \geq 0$, g, n. A path on which K and Y grow at a common constant rate exists only if $g_B = 0$: technical progress must be purely labor-augmenting.

d) (2 points, T or F) *Growth × open economy.* In a small open Solow economy with neoclassical $Y = F(K, AL)$, perfect capital mobility at the world rate r^* and residents who save a constant fraction s of national income, a permanent rise in s permanently raises GDP per worker.

Question 4. (15 points) Discursive. *Growth × national accounts × capital flows.* Brazil as a small open economy. Output is $Y = K^{\alpha}(AhL)^{1-\alpha}$, $\alpha \in (0,1)$; capital moves freely and earns the world rate $r^* > 0$ net of depreciation $\delta > 0$; A grows at $g \geq 0$ and L at $n \geq 0$, with $n + g > 0$; $h > 0$ is constant. Residents own wealth W (capital at home or abroad) and save a fraction $s \in (0,1)$ of gross national income (wages plus the gross return on wealth): $\dot{W} = s\,\mathrm{GNI} - \delta W$. Let $N \equiv n + g + \delta$; per effective worker, $k = K/(AhL)$, $y = Y/(AhL)$, $\omega = W/(AhL)$.

a) (3 points) Characterize capital and GDP per effective worker, k^* and y^*, and their dynamics. What does a rise in s do to GDP per worker? Under perfect mobility, which ratio of labor-augmenting productivity A_{BR}/A_{US} (this question's A) reconciles the gaps ϕ and χ of Question 3(a), and what does it say about the Lucas paradox?

Solution:

b) (3 points) Derive the law of motion of wealth per effective worker, its steady state ω^* and the condition for stability. For which world rates is wealth stable at every saving rate?

Solution:

c) (3 points) Assume from now on that wealth converges. Find the saving rate $\bar{s}$ above which Brazil is a net creditor in the steady state and interpret it; compare it with the Golden-Rule saving rate that Question 3(b) gives for this technology. Find GNI/GDP in the steady state in closed form.

Solution:

d) (3 points) At $t = 0$ the saving rate rises once and for all from s_0 to s_1, with $s_0 < \bar{s} < s_1$, both stable. Draw $\dot{\omega}$ against ω for both rates, marking k^* and the steady states. Find the date T at which Brazil becomes a net creditor, and the paths of GNI/GDP and GDP per worker.

Solution:

e) (3 points) *Parameter analysis.* How do ω^*/k^*, GNI/GDP, the half-life of wealth and $\bar{s}$ respond to s, r^*, $n+g$, δ and α? Prove each sign, collect the signs in a table, give the limits as $s \to 0$ and as s approaches its upper bound, and read the knife-edge cases $s = \bar{s}$ and $r^* = n + g$.

Solution:

Block 3 — Integrative: growth, welfare, national accounts and prices

(Kurlat chs. 1–2, 4–5; Romer ch. 1 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *PPP × growth (Hsieh–Klenow).* Brazil (B) and the US (U) produce a final good with $Y = K^{\alpha}(AL)^{1-\alpha}$, $\alpha \in (0,1)$, with the same A and the same $n+g+\delta > 0$; both are on their Solow balanced paths. The final good is consumed one for one or turned into new capital at a cost of p_j units of the good per unit of capital, so p_j is the price of investment relative to consumption; $p_B = \pi p_U$ with $\pi > 1$, and US prices are the international prices. At domestic prices, investment absorbs the shares $s_B, s_U \in (0,1)$ of output. Output per effective worker, in units of the final good, in Brazil relative to the US is:

A. $\left(s_B/(\pi s_U)\right)^{1/(1-\alpha)}$

B. $\left(s_B/(\pi s_U)\right)^{\alpha/(1-\alpha)}$

C. $\left(s_B/s_U\right)^{\alpha}$

D. $\left(s_B/(\pi s_U)\right)^{\alpha}$

b) (3 points) *National accounts × Golden Rule.* An economy with $Y = F(K, AL)$ (constant returns, $\tilde{y} = f(\tilde{k})$, $f' > 0 > f''$, Inada conditions) and competitive factor markets is on a balanced path on which output grows at rate $n+g > 0$. Its national accounts report a gross capital income share $\alpha_K \in (0,1)$ of GDP and a capital–output ratio $v = K/Y > 0$; capital depreciates at rate $\delta > 0$, and households discount utility at rate $\rho > 0$. Capital per effective worker is strictly below its Golden Rule level, so the path is dynamically efficient, if and only if:

A. $\alpha_K/v > n+g$

B. $\alpha_K/v - \delta > n+g$

C. $\alpha_K/v > \rho+n+g$

D. $\alpha_K/v - \delta > \rho+n+g$

c) (2 points, T or F) *Growth × welfare × discounting.* A Solow economy with $Y = K^{\alpha}(AL)^{1-\alpha}$, $\alpha \in (0,1)$, $\dot{A}/A = g$, $\dot{L}/L = n$ and $n+g+\delta > 0$ is on its balanced path with a saving rate s_0 below the Golden Rule rate s_{GR}. Current households value the path of consumption per worker by $W = \int_0^{\infty} e^{-\rho t} \ln c(t)\, dt$. A permanent rise in the saving rate at $t = 0$, to any $s_1 \in (s_0, s_{GR}]$, raises W for every discount rate $\rho > 0$.

d) (2 points, T or F) *Measurement × growth.* In the economy of item c), let the saving rate rise permanently at $t = 0$ from s_0 to any $s_1 \in (s_0, 1)$. Output per worker does not jump but consumption per worker falls; afterwards both grow faster than g, at the same rate, until the new balanced path.

Question 6. (15 points) Discursive. *Integrative: Solow transition × national accounts × welfare measurement × discounting.* Brazil debates a pension reform that would raise national saving for good. Take a closed Solow economy without government: $Y = K^{\alpha}(AL)^{1-\alpha}$, $0 < \alpha < 1$; A grows at rate g, L at rate n, capital depreciates at rate δ, and $N \equiv n+g+\delta > 0$. Per effective worker, $\tilde{y} = \tilde{k}^{\alpha}$ and $\hat{c} = (1-s)\tilde{y}$. The economy is on its balanced growth path with saving rate s_0; at $t = 0$ the reform raises the saving rate permanently to s_1, with $0 < s_0 < s_1 \leq \alpha$. A representative household values the path of consumption per person, $c(t) = A(t)\hat{c}(t)$, by $W = \int_0^{\infty} e^{-\rho t} \ln c(t)\, dt$, $\rho > 0$. Let $D(t) \equiv \ln \hat{c}_1(t) - \ln \hat{c}_0$ be the log gap between consumption per effective worker with the reform and without it ($\hat{c}_0$ is constant).

a) (3 points) Using the change of variable $z \equiv \tilde{k}^{1-\alpha}$, find the exact path of $\tilde{k}$ after the reform and its speed of convergence λ. Derive $D(t)$, its impact value $D(0)$ and its long-run value $D(\infty)$, and prove that $D(\infty) > 0$. Find the recovery time T_R at which consumption regains its pre-reform level, and the condition under which it exists.

Solution:

b) (3 points) Describe the consumption and investment shares of GDP and the growth rates of GDP and of consumption per worker along the transition. What does GDP growth say, and fail to say, about the household's welfare? Draw $D(t)$ and $\ln \tilde{y}_1(t) - \ln \tilde{y}_0$ against t, marking $D(0)$, T_R and the long-run values.

Solution:

c) (3 points) Let $\Delta W(\rho)$ be the welfare gain of the reform and $x(\rho)$ its consumption equivalent: the permanent proportional rise in the no-reform consumption path that the household values as much as the reform. Find x in terms of ΔW, show that x is strictly decreasing in ρ, find its limits as $\rho \to 0$ and $\rho \to \infty$, and prove that there is exactly one welfare-neutral discount rate ρ^*. How does ρ^* depend on N?

Solution:

d) (3 points) For a marginal reform ($s_1 \downarrow s_0$), find the sign of the derivative of $\rho\,\Delta W$ with respect to s_1 and the discount rate ρ_m at which it changes sign. Relate ρ_m to the modified golden rule of a planner who maximizes W, and prove that $\rho^* < \rho_m$ for every reform with $s_1 > s_0$.

Solution:

e) (3 points) *Parameter analysis.* Sign, with proofs, the responses of $D(0)$, $D(\infty)$, T_R, ρ^* and ρ_m to s_1, s_0 and N, and give the limits as $s_1 \downarrow s_0$, at $s_1 = \alpha$, and as $N \to 0$ or $N \to \infty$. Compare the saving rate that maximizes $D(\infty)$ with the one that maximizes the welfare change, and explain the difference. Draw $x(\rho)$ for two reforms $s_1 < s_1'$, marking where each crosses zero.

Solution: